

8.1 Magnetism

Magnet:

The material which has directional, attractive and repulsive properties and the material by which magnetic properties can be given to other materials are called magnets. Its chemical formula is Fe_3O_4 .

Magnetism:

The directional and attractive property of a magnet is called its magnetism.

Magnetic field:

The region around a magnet up to which it can show its magnetic properties is called magnetic field.

Curie point or temperature a magnet:

A particular temperature at which the magnetism of a magnet is destroyed completely is called the Curie point.

Magnetic permeability of a medium:

The characteristic of a material for which magnetic field lines can pass more easily through a material compared to another material is called magnetic permeability.

In other word, the ratio of the magnetic flux density created in a medium to the magnetic intensity is called absolute permeability of that medium. It is denoted by μ and its unit is Henry/meter (H/m).

$$\text{i.e. Magnetic permeability} = \frac{\text{Magnetic flux density}}{\text{Magnetic intensity}}; \therefore \mu = \frac{B}{H}$$

Magnetic induction:

The process by which a magnetic material becomes a magnet is called magnetic induction.

Magnetic materials and Types of them:

The materials which can be attracted and which can be transformed into magnets are called magnetic materials. For example, Iron, nickel, cobalt etc.

There are three types of magnetic materials those are given below,

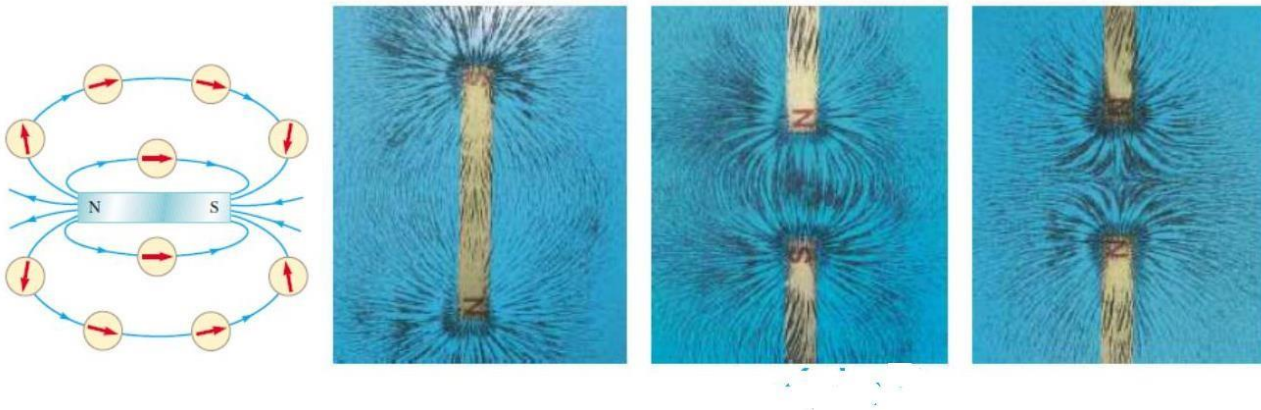
- (i) Diamagnetic substance ($\mu < 1$); Gold, Copper etc.
- (ii) Paramagnetic substance ($\mu > 1$); Platinum, Manganese etc.
- (iii) Ferromagnetic substance ($\mu \gg 1$); Iron, Cobalt etc.

MAGNETIC FIELD EFFECTS

8.1.2 Magnetic Field

Magnetic field is the region or space around any charge within which its influence can be felt by other magnetic substances.

The magnetic field around any magnet is considered as closely spaced magnetic field lines. The magnetic field lines of a bar magnet can be traced with the aid of a compass as shown in the figure below:



In addition to a bar magnet, a moving charge or a current creates a magnetic field in the surrounding space (in addition to its electric field). The magnetic field exerts a force on any other moving charge or current that is present in the field.

8.2 Magnetic Force on a Charged Particle

Consider a charged object having charge q is projected a uniform magnetic field of flux density with velocity $\mathbf{v}$. The magnetic force acting on the object can be expressed as:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

$$F = qvB \sin \theta \quad \hat{n} \text{ --- (1)}$$

Here $\hat{n}$ is the unit vector that is perpendicular to the plane of $\mathbf{v}$ and $\mathbf{B}$, and is used to describe the direction of magnetic force on the charged object.

It is clear from equation (1) that the maximum magnetic force will act on the charged object when it will be projected perpendicular to the magnetic field. The maximum magnetic force on the charged object will be

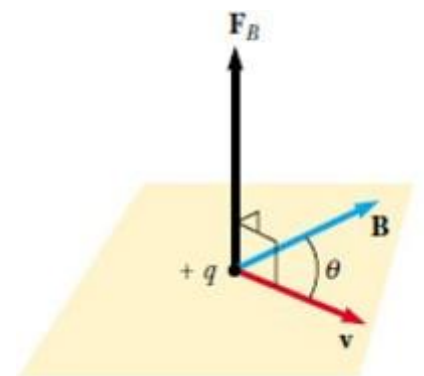
$$F = qvB$$

$$B = \frac{F}{qv}$$

$\mathbf{B}$ is also called the magnetic field induction which can be defined as:

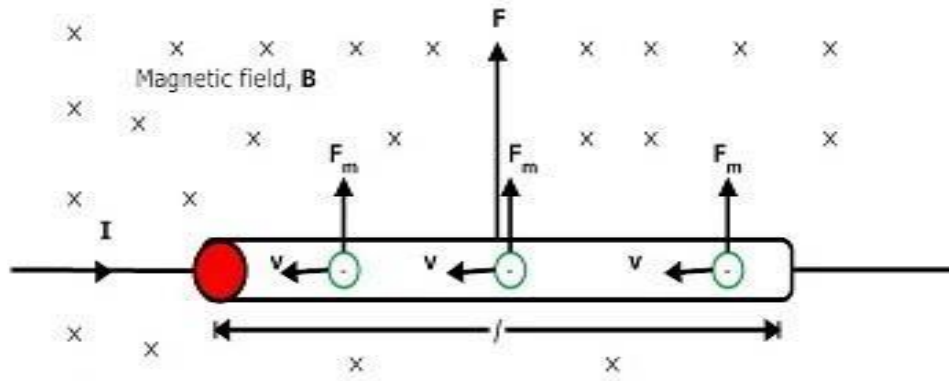
“The force on a unit positive charge moving perpendicular to the magnetic field with uniform velocity”. The SI unit of magnetic field induction $\mathbf{B}$ is tesla, while in cgs system of units $\mathbf{B}$ is measured in gauss.

$$1 \text{ tesla} = 10^4 \text{ gauss}$$



8.3 Magnetic Force on a Current Carrying Conductor

Consider a current carrying wire of length L and cross-sectional area A , is placed in a uniform magnetic field of flux density $\mathbf{B}$ as shown in the figure below:



If n is the number of free charges per unit volume of a conductor (each having charge e), then the total charge flowing through the wire is

$$q = nAle$$

Suppose the charges are moving with drift velocity v_d and cover length l in t seconds, then

$$t = \frac{l}{v_d}$$

If I is the current flowing through the conductor, then

$$I = \frac{q}{t}$$

$$\Rightarrow I = \frac{nAle}{\frac{l}{v_d}}$$

$$\therefore v_d = \frac{I}{nAe}$$

Now

$$\text{Force on one charge } F' = ev_d B_{\perp}$$

$$\text{Force on } nAl \text{ charges, } F = nAlF' = nAle v_d B_{\perp}$$

Putting value of v_d we get,

$$F = nAle \left(\frac{I}{nAe} \right) B_{\perp} = IB_{\perp} l$$

In vector form:

$$\vec{F} = I(\vec{l} \times \vec{B})$$

The direction of F will be perpendicular to the plane of $\vec{l}$ and $\vec{B}$.

8.3.1 Torque on a Current Loop in Magnetic Field

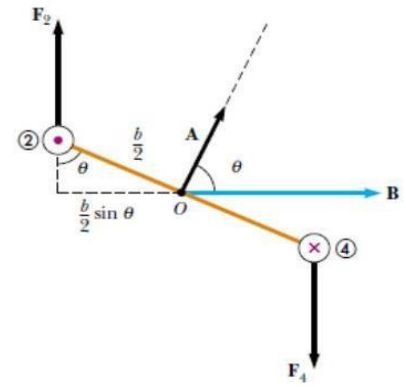
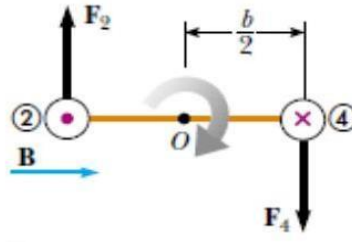
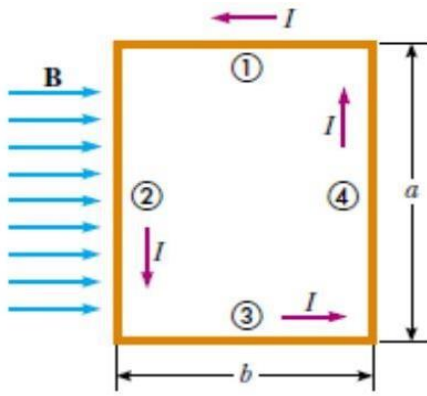
Consider a loop of wire carrying current suspended in a uniform magnetic field of flux density $\mathbf{B}$ as shown in the figure below:

The sides 1 and 3 of the coils are oriented parallel to the magnetic field. Thus,

$$F_1 = IbB \sin 180^\circ = 0 \text{ and}$$

$$F_3 = IbB \sin 0^\circ = 0$$

The sides 2 and 4 are held perpendicular to the magnetic field. Thus,



$$F_2 = F_4 = F = IlB\sin\theta = IaB\sin 90^\circ = IaB \quad \because l = a$$

The direction of F_2 , is out of the page and that of F_4 , is into the page. These two anti-parallel forces separated by a small distance form a couple. The couple tends to rotate the loop about point O. The magnitude of torque of this couple will be:

Torque = (Force)(Perpendicular distance betⁿ line of action of Force)

$$\tau = Fb\sin\theta = IaB b\sin\theta = IabB\sin\theta$$

Where θ is the angle between the magnetic field $\mathbf{B}$ and vector area $\mathbf{A}$ of the loop.

$$\tau = IAB\sin\theta \because A = ab = \text{area of loop}$$

This is the expression of torque produced in a loop with single turn. For a loop with N turns, the torque will be:

$$\tau = NIAB\sin\theta$$

In vector form:

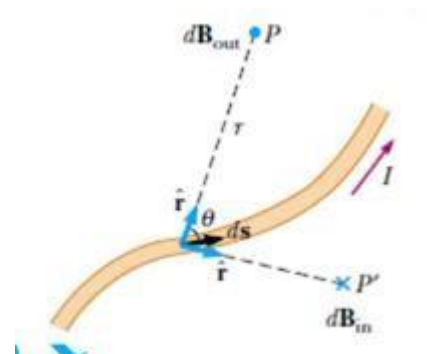
$$\boldsymbol{\tau} = NI(\mathbf{A} \times \mathbf{B})$$

This is the expression of the torque on a current carrying loop in a magnetic field.

8.4 Biot-Savart's law:

Statement: When current flows through a conductor, magnetic field is generated around it. According to this law, the magnitude of magnetic field produced around a conductor of small length due to the flow of current through it is –

- Directly proportional to the amount of **current**,
- Directly proportional to the **length** of the conductor,
- Directly proportional to the **sine of the angle** between the lines connecting the point from the mid-point of the conductor and the conductor itself,
- Inversely proportional to the **square of the distance** between the point and the mid-point of the conductor.



Explanation:

According to this law, the magnetic flux density dB at a point P due to a small element dl of a conductor carrying a current I is given by,

$$dB \propto \frac{I dl \sin\theta}{r^2} \dots \dots \dots (1)$$

Where, r is the distance from the point P to the mid-point of the element dl and θ is the angle between the element and the line joining it to P as shown in Fig-3.

From equation (1) we can write,

$$dB = K \frac{I dl \sin\theta}{r^2} \dots \dots \dots (2)$$

Where, K is a proportionality constant whose value depends on nature of the medium and the unit of the quantities.

For air medium and in S.I. unit

$$K = \frac{\mu_0}{4\pi}$$

Where, $\mu_0 = 4\pi \times 10^{-7} \text{H/m}$.

So, putting this value in (2) we get,

$$dB = \frac{\mu_0 I \cdot dl \cdot \sin\theta}{4\pi r^2}$$

For other medium we get,

$$dB = \frac{\mu}{4\pi} \frac{I \cdot dl \cdot \sin\theta}{r^2}$$

where, μ is the permeability of that medium.

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

9.1 Magnetic Flux

The number of magnetic lines of force passing normally through certain area is called magnetic flux. It is denoted by ϕ_B . It is a scalar quantity and its SI unit is weber (Wb). It is measured by the product of magnetic field strength and the component of vector area parallel to magnetic field.

If $d\mathbf{A}$ is the vector area element of the surface placed in uniform magnetic field of magnetic field strength B as shown figure below.

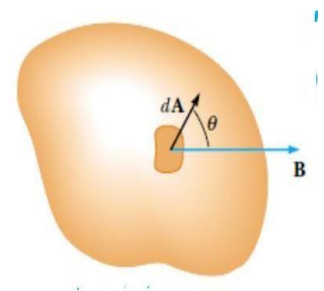
The magnetic flux $d\phi_B$ through $d\mathbf{A}$ is given by:

$$d\phi_B = \mathbf{B} \cdot d\mathbf{A}$$

$$\phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

$$\phi_B = \int B dA \cos\theta$$

Where θ is the angle between magnetic field strength and vector the area element.



9.2 Faraday's Law of Electromagnetic Induction

Statement:

The induced emf in a circuit is equal to the negative of rate at which the magnetic flux through the circuit is changing with time.

Or

The magnitude of induced emf in a circuit is directly proportional to the rate of change of magnetic flux.

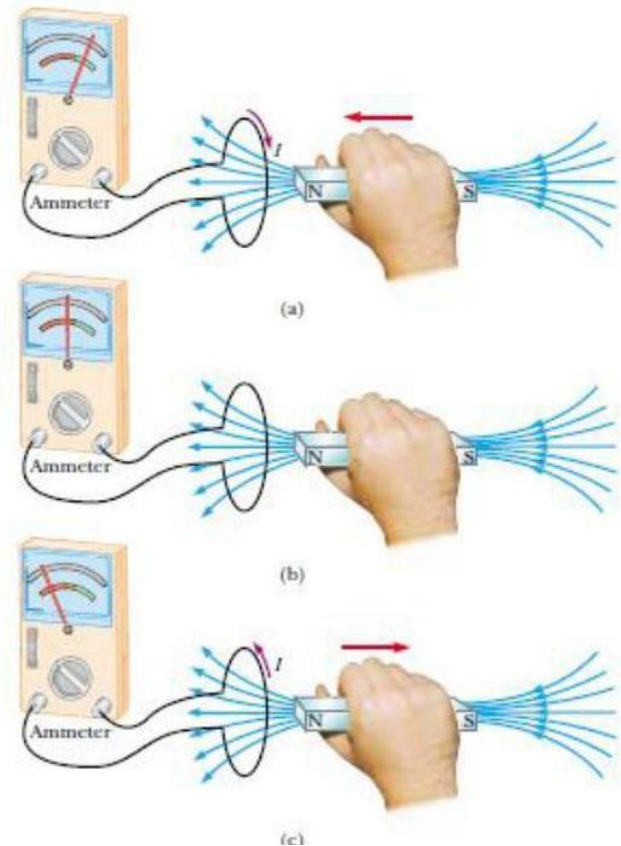
*Explanation

When a magnet is moved toward the loop, the ammeter needle deflects in one direction, as shown in the figure (a). When the magnet is brought to rest and held stationary relative to the loop figure (b), no deflection is observed. When the magnet is moved away from the loop, the needle deflects in the opposite direction, as shown in figure

(c). Finally, if the magnet is held stationary and the loop is moved either toward or away from it, the needle deflects.

From these observations, we conclude that the loop detects that the magnet is moving relative to it and we relate this detection to a change in magnetic field. Thus, it seems that a relationship exists between current and changing magnetic field.

These results are quite remarkable in view of the fact that a current is set up even though no batteries are present in the circuit. We call such a current the induced current which is produced by an induced emf . This phenomenon is called electromagnetic induction.



Mathematical form of Faraday's Law

If $\mathcal{E}$ is the induced emf due to change in flux $d\phi_B$ in time dt , then we can describe the Faraday's law of electromagnetic induction as:

$$\mathcal{E} = - \frac{d\phi_B}{dt}$$

If there are N number of turns in a coil, then,

$$\mathcal{E} = - \frac{d(N\phi_B)}{dt}$$

$$\mathcal{E} = -N \frac{d\phi_B}{dt}$$

9.3 Lenz law

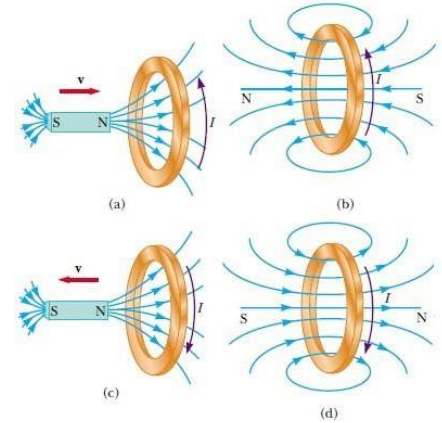
The direction of induced current is such that it opposes its own cause.

Explanation:

Consider a bar magnet moves toward a stationary metal loop, as in figure (a). As the magnet moves to the right toward the loop, the external magnetic flux through the loop increases with time. As the result, the induced current set up in the loop which produces magnetic field, as illustrated in figure (b).

Knowing that like magnetic poles repel each other, we conclude that the left face of the current loop acts like a north pole and that the right face acts like a south pole.

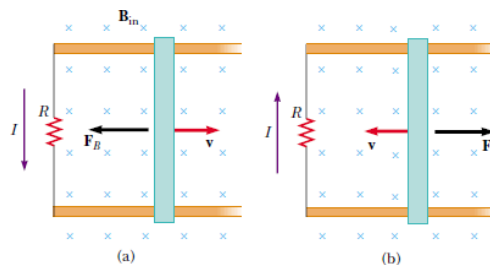
If the magnet moves to the left, as in figure (c), its flux through the area enclosed by the loop decreases in time. Now the induced current in the loop produces the magnetic field as shown in figure (d). In this case, the left face of the loop is a south pole and the right face is a north pole.



9.3.1 Lenz Law and Conservation of Energy

*“Lenz law is the statement of law of conservation of energy
for the circuit involving induced current”*

To understand this statement, consider a conducting bar moving to the right on two parallel rails in the presence of a uniform magnetic field as shown in the figure below. As the bar moves to the right, the magnetic flux through the area enclosed by the circuit increases with time because the area increases. As the result, the induced current must be directed counterclockwise when the bar moves to the right. Since the current carrying bar is moving in the magnetic field, it will experience a magnetic force. By using right hand rule, the direction of is opposite to that of, that tends to stop the rod. An external dragging force must be applied to keep the rod moving in the magnetic field.

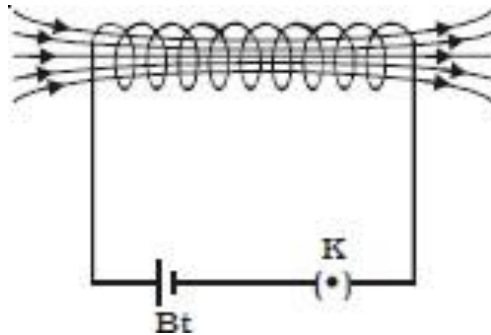


This dragging force provides the energy for the induced currents to flow. This energy is the source of induced current. Thus, the electromagnetic induction is exactly according to the law of conservation of energy.

If the bar is moving to the left, as in figure (b), the external magnetic flux through the area enclosed by the loop decreases with time. Because the field is directed into the page, the direction of the induced current must be clockwise.

9.4 Self-Induction

The property of a coil which enables to produce an opposing induced *emf* in it when the current in the coil changes is called self-induction.



Coefficient of self induction

When a current I flows through a coil, the magnetic flux (ϕ) linked with the coil is proportional to the current.

$$\phi \propto I$$

$$\phi = LI$$

Where L is a constant of proportionality and is called coefficient of self induction or self inductance.

From Faraday's law of induction.

$$\mathcal{E} = -\frac{d\phi}{dt}$$

So,

$$\mathcal{E} = -\frac{d(LI)}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

If there are N no. of turns in the coil then,

$$\mathcal{E} = -N \frac{dI}{dt}$$

The unit of self inductance is henry.

9.5 Mutual induction

The phenomenon of producing an induced *emf* in a coil due to the change in current in the other coil is known as mutual induction.

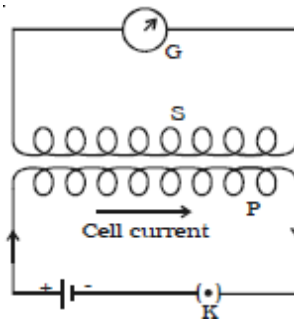


Figure: Mutual Induction

Coefficient of mutual induction

When a current I flows through a coil, the magnetic flux (ϕ) linked with the other coil is proportional to the current.

$$\phi \propto I$$

$$\phi = MI$$

Where M is a constant of proportionality and is called coefficient of mutual induction or mutual inductance.

From Faraday's law of induction.

$$\mathcal{E} = -\frac{d\phi}{dt}$$

So,

$$\mathcal{E} = -\frac{d(MI)}{dt}$$

$$\mathcal{E} = M \frac{dI}{dt}$$

If there are N no. of turns in the coil then,

$$\mathcal{E} = -NM \frac{dI}{dt}$$

The unit of mutual inductance is henry.

Magnetic Properties of Matter

Factors Determining Magnetic Properties

The most basic factor is the configuration of the electrons in the material's atoms. The ability of the atoms or molecules in the material to align magnetically is important in determining whether the material responds to a magnetic field. The orientation of the spin of the electrons in an atom, the orientation of the atoms in a molecule or alloy, and the ability of domains of atoms or molecules to line up are the factors that determine how a material responds to a magnetic field.

Electron orientation

Electrons can behave as tiny magnets, each with north (N) and south (S) poles. When an atom's electrons are lined up in the same orientation, with most having their N pole facing one direction, the atom becomes like a magnet, with N and S poles. It is also possible for the electrons to be in various directions, making the atom not magnetic.

Moving electrons create magnetic field

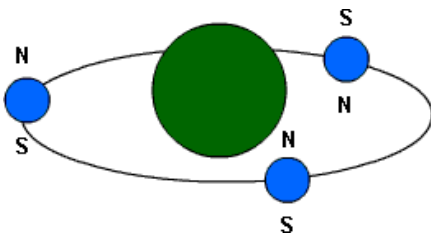
The reason that electrons can behave like tiny magnets is the fact that when electrons move, they create a magnetic field. Placing a compass near a wire carrying DC electrical current can show that a magnetic field is created due to the electrons moving through the wire.

A magnetic field is also created when electrons rotate around a nucleus and when they spin while in orbit.

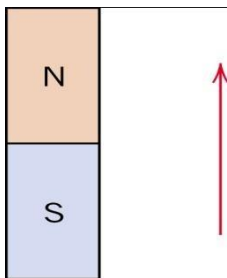
Spinning electrons

Electrons have a property called *spin*. This spinning creates a magnetic field with N and S poles; just as the spinning Earth has magnetic poles. Note that the N pole on an electron is really a North-seeking pole, just as in a magnet.

If electrons in the shells of an atom spin in the same direction, the atom will exhibit a magnetic field and will respond to the forces of a magnet. If half of the electrons spin one way and the rest spin the other way, they will neutralize each other and the material will not be affected by a magnetic field. This atom is barely magnetic because all its electrons are not aligned.



The magnetic moment represented by a vector



Strong and weak electron alignments

Atoms such as iron have most of their electrons aligned in the same direction. Thus, iron or nickel would be attracted to a magnet. Aluminum only has a few electrons aligned, and thus it is only weakly magnetic. An element with half of its electrons oriented one way would not be attracted to a magnet.

Atomic and molecular alignment

Although some atoms may be highly magnetic, they really need to be aligned to make a material magnetic. If magnetic atoms are facing different directions, their fields will cancel out each other.

Solids and fluids

Since the atoms or molecules in a solid are fixed in place, most magnetic materials are solids. This is because once the atoms or molecules become aligned, they tend to stay in place. An example is seen when you magnetize a piece of iron.

As a material becomes heated or when it is in its liquid or gaseous state, the atoms or molecules are in rapid motion and are not aligned. Thus, fluids are seldom magnetic.

An exception is when a magnetic material such as iron is in its liquid state and is continually rotating around an axis. In such a situation, the atoms can be aligned in one direction, even though they are in rapid motion.

For example, the core of the Earth is made of liquid iron. Since the Earth rotates on its axis, the liquid iron is rotating, thus creating the Earth's magnetic field. Also, the Sun rotates on its axis, and the material in its plasma state creates the Sun's magnetic field.

Molecules

If two or more elements are chemically combined to form a molecule, it is quite possible that the compound is not very magnetic because the orientations of the atoms in the molecule work against each other.

A good example of this is to compare the magnetic properties of iron as compared to its compounds if iron oxide (rust) and iron sulfide. A piece of iron is highly magnetic, but a hunk of rusty iron is not.

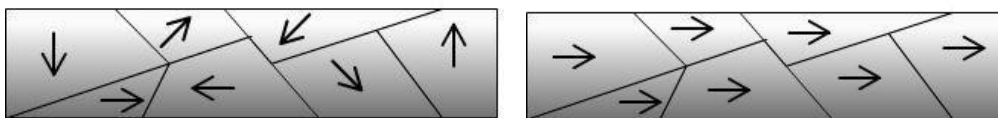
Alloys

Metals of different elements can be mixed when they are in the molten or liquid state to form alloys. These combinations result in materials with slightly different physical and chemical properties than the elements by themselves.

If the metals typically respond well to a magnetic field--such as iron and nickel--then their alloy has even a stronger reaction to magnetism. On the other hand, there are some alloys of iron--such as forms of stainless steel--that do not respond well at all to a magnet.

Domains

The final factor is the orientation of domain in a solid. A group of atoms in a metal may become aligned, but the various groups may be misaligned. These groups are called domains. It is necessary to line up many of the domains in a material like iron in order for it to become a magnet.



Magnetic material with domains misaligned, aligned domains make material highly magnetic

Summary

Alignment of electrons, atoms and domains are important in determining the magnetic response of a material and whether it is a magnet. Since the atoms or molecules need to be aligned, gases and liquids are typically not magnetic, and most magnets are solid metals. An exception is in the rotating liquid iron core of the Earth and the rotating plasmas of the Sun.

Magnetic susceptibility

Magnetic susceptibility is a dimensionless proportionality constant that indicates the degree of magnetization of a material in response to an applied **magnetic field**.

Magnetic permeability

In electromagnetism, permeability is the measure of the ability of a material to support the formation of a magnetic field within itself. Hence, it is the degree of magnetization that a material obtains in response to an applied magnetic field.

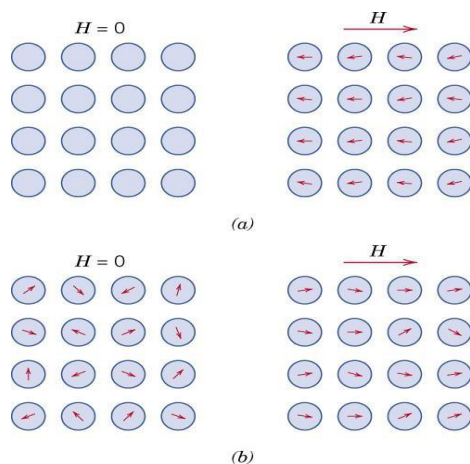
Diamagnetic, Paramagnetic, and Ferromagnetic Materials

When a material is placed within a magnetic field, the magnetic forces of the material's electrons will be affected. This effect is known as Faraday's Law of Magnetic Induction. However, materials can react quite differently to the presence of an external magnetic field. This reaction is dependent on a number of factors, such as the atomic and molecular structure of the material, and the net magnetic field associated with the atoms. The magnetic moments associated with atoms have three origins. These are the electron motion, the change in motion caused by an external magnetic field, and the spin of the electrons.

In most atoms, electrons occur in pairs. Electrons in a pair spin in opposite directions. So, when electrons are paired together, their opposite spins cause their magnetic fields to cancel each other. Therefore, no net magnetic field exists. Alternately, materials with some unpaired electrons will have a net magnetic field and will react more to an external field. Most materials can be classified as diamagnetic, paramagnetic or ferromagnetic.

Diamagnetic

- Materials have a weak, negative susceptibility to magnetic fields.
- Repelled by a magnetic field
- The material does not retain the magnetic properties when the external field is removed.
- No hysteresis properties, curie points and retentivity.
- In diamagnetic materials all the electron are paired so there is no net permanent magnetic moment per atom.
- Diamagnetic properties arise from the realignment of the electron paths under the influence of an external magnetic field.
- The value of diamagnetic susceptibility is independent of temperature and applied magnetic field.
- Most elements in the periodic table, including copper, silver, and gold, Bismuth, antimony, Zn, organic materials, Alkali earth metals are diamagnetic. They can be solids, liquids and gases.



Paramagnetic

- Materials have a small, positive susceptibility but of low order to magnetic fields.

- Slightly attracted by a magnetic field and the material do not retain the magnetic properties when the external field is removed.
- Paramagnetic properties are due to the presence of some unpaired electrons, and from the realignment of the electron paths caused by the external magnetic field
- No hysteresis properties, curie points and retentivity.
- Paramagnetic substance tends to go slowly to the stronger region of the field from the weaker region.
- Paramagnetic materials include magnesium, molybdenum, lithium, and tantalum. They can be solids, liquids and gases.
- Paramagnetic materials like aluminum, uranium and platinum become more magnetic when they are very cold.

Ferromagnetic

- Materials have a large, positive susceptibility to an external magnetic field.
- Exhibit a strong attraction to magnetic fields and are able to retain their magnetic properties after the external field has been removed.
- They have fixed curie points
- Show hysteresis properties.
- Magnetic permeability is of high order.
- Their susceptibility depends on temperature i.e., $K \propto \frac{1}{T}$
- Ferromagnetic substance tends to go quickly to stronger region of the field from weaker region.
- Ferromagnetic materials have some unpaired electrons so their atoms have a net magnetic moment.
- They get their strong magnetic properties due to the presence of magnetic domains. In these domains, large numbers of atom's moments (10^{12} to 10^{15}) are aligned parallel so that the magnetic force within the domain is strong.
- When a ferromagnetic material is in the unmagnetized state, the domains are nearly randomly organized and the net magnetic field for the part as a whole is zero. When a magnetizing force is applied, the domains become aligned to produce a strong magnetic field within the part.
- Iron, nickel, and cobalt are examples of ferromagnetic materials. They can be crystalline and solids.

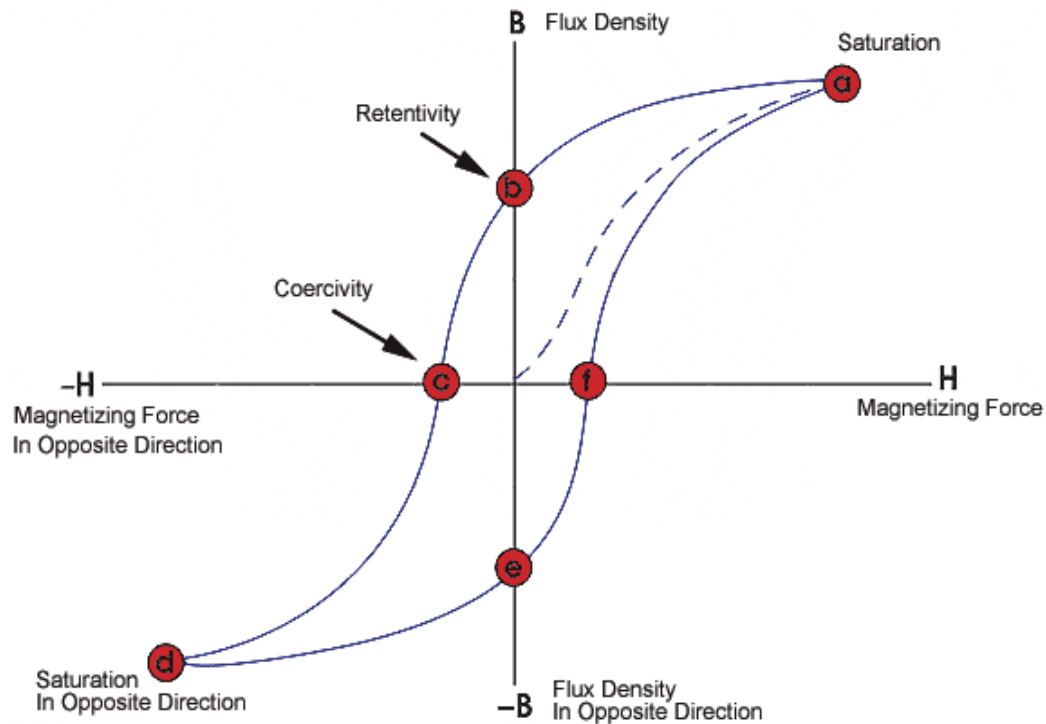
Temperature effect

Strongly magnetic ferromagnetic materials like nickel or steel lose all their magnetic properties if they are heated to a high enough temperature. The atoms become too excited by the heat to remain pointing in one direction for long.

The temperature at which a metal loses its magnetism is called the Curie temperature, and it is different for every metal. The Curie temperature for nickel, for example, is about 350°C.

The Hysteresis Loop and Magnetic Properties

“The striking property of Ferro Magnetic materials is the relation between Magnetization and the strength of Magnetic field. This property is called Hysteresis”



The loop is generated by measuring the magnetic flux of a ferromagnetic material while the magnetizing force is changed. A ferromagnetic material that has never been previously magnetized or has been thoroughly demagnetized will follow the dashed line as H is increased.

As the line demonstrates, the greater the amount of current applied ($H+$), the stronger the magnetic field in the component ($B+$). At point "a" almost all of the magnetic domains are aligned and an additional increase in the magnetizing force will produce very little increase in magnetic flux. The material has reached the point of magnetic saturation.

When H is reduced to zero, the curve will move from point "a" to point "b." At this point, it can be seen that some magnetic flux remains in the material even though the magnetizing force is zero. This is referred to as the point of retentivity on the graph and indicates the remanence or level of residual magnetism in the material. (Some of the magnetic domains remain aligned but some have lost their alignment.)

As the magnetizing force is reversed, the curve moves to point "c", where the flux has been reduced to zero. This is called the point of coercivity on the curve. (The reversed magnetizing force has flipped enough of the domains so that the net flux within the material is zero.) The force required to remove the residual magnetism from the material is called the coercive force or coercivity of the material.

As the magnetizing force is increased in the negative direction, the material will again become magnetically saturated but in the opposite direction (point "d"). Reducing H to zero brings the curve to point "e." It will have a level of residual magnetism equal to that achieved in the other direction. Increasing H back in the positive direction will return B to zero. Notice that the curve did not return to the origin of the graph because some force is required to remove the residual magnetism. The curve will take a different path from point "f" back to the saturation point where it will complete the loop. The area of loop indicates the amount of energy wasted in one cycle of operation.

From the hysteresis loop, a number of primary magnetic properties of a material can be determined.

1. **Retentivity** - A measure of the residual flux density corresponding to the saturation induction of a magnetic material. In other words, it is a material's ability to retain a certain amount of residual magnetic field when the magnetizing force is removed after achieving saturation. (The value of B at point b on the hysteresis curve.)
2. **Residual Magnetism** or **Residual Flux** - the magnetic flux density that remains in a material when the magnetizing force is zero. Note that residual magnetism and retentivity are the same when the material has been magnetized to

the saturation point. However, the level of residual magnetism may be lower than the retentivity value when the magnetizing force did not reach the saturation level.

3. **Coercive Force** - The amount of reverse magnetic field which must be applied to a magnetic material to make the magnetic flux return to zero. (The value of H at point c on the hysteresis curve.)
4. **Permeability, μ** - A property of a material that describes the ease with which a magnetic flux is established in the component.
5. **Reluctance** - Is the opposition that a ferromagnetic material shows to the establishment of a magnetic field. Reluctance is analogous to the resistance in an electrical circuit.

Hard Magnetic Materials

- ✓ Hard magnetic materials have large hysteresis loss due to large hysteresis loop area
- ✓ The Coercivity and retentivity are large hence these materials cannot be easily magnetized and demagnetized.
- ✓ These materials have small values of permeability and susceptibility.
- ✓ These are used to make permanent magnets.
- ✓ Ex. Alnico alloy

Soft Magnetic Materials

- ✓ Soft Magnetic materials have low hysteresis loss due to small hysteresis loop area.
- ✓ The coercivity and retentivity are small; hence these materials can be easily magnetized and demagnetized.
- ✓ These materials have large values of permeability and susceptibility.
- ✓ These are used to make electromagnets.
- ✓ Ex: Iron silicon alloys, Ferrous nickel alloy

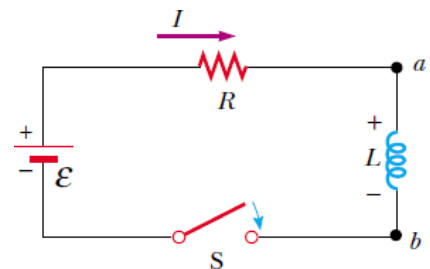
10.1 Energy Stored in the Magnetic Field

Consider a resistor and an inductor is connected in series with a battery of emf $\mathcal{E}$. According to the Kirchhoff's 2nd rule;

$$\mathcal{E} - V_R - V_L =$$

$$0 \quad \mathcal{E} = V_R + V_L$$

$$\mathcal{E} = IR + L \frac{dI}{dt}$$



Multiplying both sides by I , we have:

$$\mathcal{E}I = I^2R + LI \frac{dI}{dt}$$

Here $\mathcal{E}I$ is the power supplied by the source, I^2R is the power dissipated in resistance and $LI \frac{dI}{dt}$ is the energy supplied per time in the inductance coil, where the magnetic field also exists.

If U_B is the total energy stored in the magnetic field of inductance coil, then the energy stored per unit time is:

$$\frac{dU_B}{dt} = LI \frac{dI}{dt}$$

$$dU_B = LI dI$$

Integrating both sides, we get:

$$\int_0^{U_B} dU_B = \int_0^I LI \, dI$$

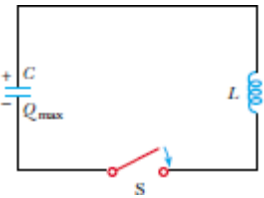
$$U_B = L \left[\frac{I^2}{2} \right]_0^I$$

$$U_B = \frac{1}{2} LI^2$$

This is the expression of energy stored in the magnetic field of a current carrying inductor.

10.2.1 Electromagnetic Oscillations (Qualitative discussion)

Consider an oscillating circuit which consists of a capacitor and an inductor. If the capacitor is initially charged and the switch is then closed, we find that both the current in the circuit and the charge on the capacitor oscillate between maximum positive and negative values. If the resistance of the circuit is zero, no energy is transformed to internal energy. In the following analysis, we neglect the resistance in the circuit. We also assume an idealized situation in which energy is not radiated away from the circuit.



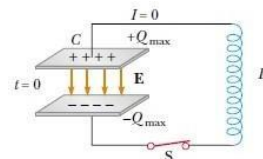
When the capacitor is fully charged, the energy in the circuit is stored in the electric field of the capacitor and is equal to

$$U_E = \frac{1}{2} \frac{Q_{\max}^2}{C}$$

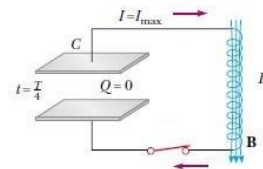
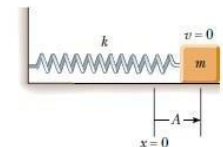
At this time, the current in the circuit is zero, and therefore no energy is stored in the inductor.

$$U_B = 0$$

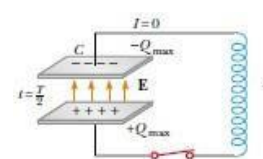
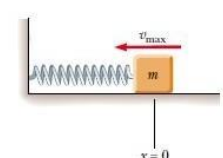
As the switch S is closed, the capacitor begins to discharge and the energy stored in its electric field decreases. The discharge of the capacitor represents a current in the circuit, and hence some energy is now stored in the magnetic field of the inductor. Thus, energy is transferred from the electric field of the capacitor to the magnetic field of the inductor. When the capacitor is fully discharged, it stores no energy. At this time, the current reaches its maximum value, and all of the energy is stored in the inductor.



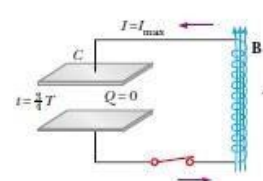
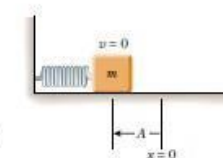
(a)



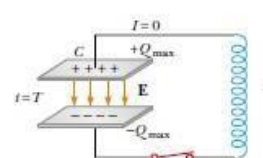
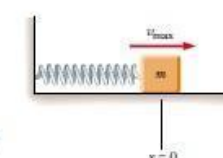
(b)



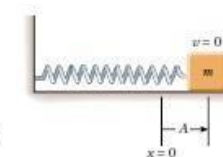
(c)



(d)

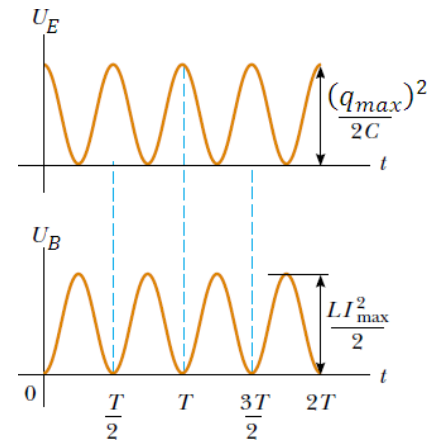


(e)



$$(U)_E \min = 0, (U)_B \max = \frac{1}{2} L I_{\max}^2$$

The current continues in the same direction, decreasing in magnitude, with the capacitor eventually becoming fully charged again but with the polarity of its plates now opposite the initial polarity. This is followed by another discharge until the circuit returns to its original state of maximum charge $q_{\max}$. The energy continues to oscillate between inductor and capacitor. When the energy stored in the capacitor is maximum, the energy stored in inductor is minimum and vice versa. But the total energy $U = U_E + U_B$ remains constant.



As the energy in such circuit is used to oscillate between capacitor and inductor, therefore, it is also called ‘Oscillating Circuit’.

10.2.2 LC Circuit: Electromagnetic Oscillations (Quantitative analysis)

Let, U_E and U_B are the values of energy stored in the electric field of capacitor and magnetic field of an inductor, respectively.

So, the total energy U stored in the circuit is:

$$U = U_E + U_B$$

$$U = \frac{1}{2} \frac{q^2}{C} + \frac{1}{2} L I^2$$

Differentiating the above equation we get,

$$\frac{dU}{dt} = \frac{d}{dt} \left[\frac{1}{2} \frac{q^2}{C} + \frac{1}{2} L I^2 \right]$$

as, $U = \text{constant}$, we have,

$$0 = \frac{q}{C} \frac{dq}{dt} + L I \frac{dI}{dt}$$

$$\Rightarrow \frac{q}{C} \frac{dq}{dt} + L \frac{dq}{dt} \frac{d}{dt} \left(\frac{dq}{dt} \right) \quad [\because I = \frac{dq}{dt}]$$

$$\Rightarrow \frac{dq}{dt} \left[\frac{q}{C} + L \frac{d^2 q}{dt^2} \right] = 0$$

$$\Rightarrow \frac{q}{C} + L \frac{d^2 q}{dt^2} = 0$$

$$\therefore \frac{d^2 q}{dt^2} + \frac{1}{LC} q = 0 \dots \dots \dots (1)$$

Solution of equation (1) will be,

$$q = q_m \cos(\omega t + \phi) \dots \dots \dots (2)$$

Where,

$$\omega = \frac{1}{\sqrt{LC}}$$

Thus, time period, $T = \frac{2\pi}{\omega}$

$$T = 2\pi\sqrt{LC}$$

And frequency,

$$f = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

Total Energy in LC Oscillations:

The energy stored in the electric field of a capacitor is,

$$U_E = \frac{1}{2} \frac{q^2}{C}$$

$$U_E = \frac{1}{2C} q_m^2 \cos^2(\omega t + \phi) \dots \dots \dots (3)$$

The energy stored in the magnetic field of an inductor will be,

$$U_B = \frac{1}{2} LI^2$$

$$U_B = \frac{1}{2} L \left(\frac{dq}{dt} \right)^2$$

$$U_B = \frac{1}{2} L [\omega^2 q_m^2 \sin^2(\omega t + \phi)]$$

$$U_B = \frac{1}{2} L \left[\frac{1}{LC} q_m^2 \sin^2(\omega t + \phi) \right]$$

$$U_B = \frac{1}{2C} q_m^2 \sin^2(\omega t + \phi) \dots \dots \dots (4)$$

The total energy stored in the LC circuit is:

$$U = U_E + U_B$$

$$U = \frac{1}{2C} q_m^2 \cos^2(\omega t + \phi) + \frac{1}{2C} q_m^2 \sin^2(\omega t + \phi)$$

$$U = \frac{1}{2C} q_m^2 [\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)]$$

$$U = \frac{1}{2C} q_m^2$$

As, q_m is the maximum charge stored in the capacitor, so the total energy stored in LC circuit is constant.